



Tecnológico de Monterrey
Partial Exam 1 A
2026-Jan-May

Grade

Course: Calculus I

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 18/02/2026

Last Names and Names: _____

Group: 201

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Multiple option section. Read the following questions and identify the option that best answers each, then write its letter on the line.

1. (5 points) _____ Select the option that is the definition of the derivative:

(a) $\lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{h}$ (b) $\lim_{h \rightarrow 0} \frac{f(x-h) + f(x)}{h}$ (c) $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

2. (5 points) _____ Identify the quotient rule for derivatives.

(a) $\left(\frac{u}{v}\right)' = \frac{uv' - u'v}{v^2}$ (b) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ (c) $\left(\frac{m}{n}\right)' = \frac{mn' - n'm}{n^2}$ (d) $\left(\frac{m}{n}\right)' = \frac{n'm - m'n}{n^2}$

3. (5 points) _____ Identify the product rule for derivatives

(a) $(uv)' = u'v + uv'$ (b) $(uv)' = u'v + u'v$ (c) $(uv)' = uv' + uv'$ (d) $(uv)' = u' + v'$

4. (5 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} e^x = xe^x$ (b) $\frac{d}{dx} e^x = e^x$ (c) $\frac{d}{dx} e^x = xe^{x-1}$ (d) $\frac{d}{dx} e^x = \frac{1}{x}e^x$

5. (5 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} \cos(x) = \cos(x)$ (b) $\frac{d}{dx} \cos(x) = -\sin(x)$ (c) $\frac{d}{dx} \cos(x) = \sin(x)$ (d) $\frac{d}{dx} \cos(x) = -\cos(x)$

6. (5 points) _____ Which of the following derivatives is correct

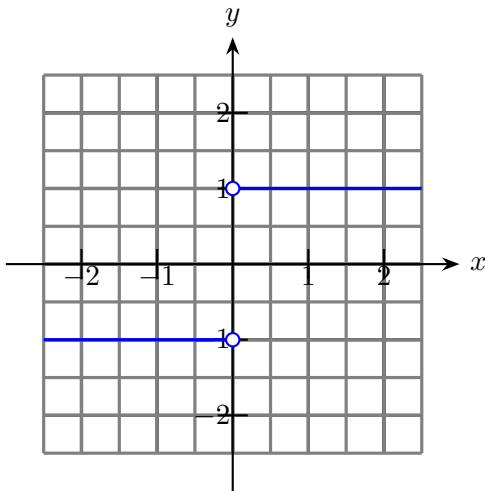
(a) $\frac{d}{dx} \sin(x) = \cos(x)$ (b) $\frac{d}{dx} \sin(x) = -\sin(x)$ (c) $\frac{d}{dx} \sin(x) = \sin(x)$ (d) $\frac{d}{dx} \sin(x) = -\cos(x)$

7. (5 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} \ln(x) = \frac{1}{\ln(x)}$ (b) $\frac{d}{dx} \ln(x) = \frac{1}{x}$ (c) $\frac{d}{dx} \ln(x) = \frac{\ln(x)}{x}$ (d) $\frac{d}{dx} \ln(x) = \frac{1}{x \ln(x)}$

II Limits. Properties, Numerical, Graphic and Algebraic method.

1. (5 points) Applying the graphical technique to compute the limit when $x \rightarrow 0$. Argument if the limit converges. Use the notion of limit from the left and right.



2. (5 points) Compute the limit of the function $f(x) = \frac{6x^2+4}{5x^2-2}$ as $x \rightarrow \infty$ using the algebraic method.

3. (5 points) Compute the limit of the function $f(x) = \frac{\sqrt{x+9}-3}{x}$ as $x \rightarrow 0$ using the numerical method.

III Derivatives by definition.

1. (13 **points**) Compute the derivative of the function $f(x) = \frac{x^2-5x+6}{x-2}$ using the definition of derivatives at $x = 3$.

IV Derivatives by rules. Use the power, product and quotient rules to compute the derivatives for the given functions.

1. (12 **points**) Compute the derivative of the function $f(x) = 4 \left[\left(\frac{0.5}{x} \right)^4 - 1 \right]$.

2. (12 points) Compute the derivative of the function $g(x) = \frac{e^x + \sin(x)}{x^2}$.

3. (13 points) Compute the derivative of the function $f(x) = x \cdot \ln(x) \cdot e^x$.